Reinforcement Learning for Beer Game Inventory Control

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Abstract

This report studies single-agent reinforcement learning for inventory ordering in the Beer Game. We implement and compare DQN, Double DQN, and PPO in the standard scalar order space, and High-Dim DQN, High-Dim Double DQN, and High-Dim PPO in a three-dimensional vector order space. In the complete single-run experiments, standard-action PPO obtains the highest test reward, 627.90, with a demand satisfaction rate of 0.934. In the high-dimensional setting, High-Dim PPO obtains the best single-run reward, 406.63, and outperforms High-Dim DQN and High-Dim Double DQN. To reduce the effect of single-run randomness and mismatched evaluation streams, we further run a controlled high-dimensional experiment with three shared seed offsets and reset evaluation randomness before testing. In this controlled comparison, High-Dim PPO reaches 407.18 ± 21.33 mean reward across seeds, outperforming High-Dim DQN at 339.91 ± 18.37 and High-Dim Double DQN at 306.04 ± 46.72; the best High-Dim PPO run reaches 424.95. Based on reward, stability, and interpretability, we select High-Dim PPO as the main method for the high-dimensional order-space experiment.

1. Introduction

The Beer Game is a classical dynamic decision problem in supply-chain management. Firms are arranged in a serial chain: downstream firms face stochastic customer demand, while upstream firms receive orders from downstream firms. Ordering decisions affect holding cost, shortage penalties, and procurement cost, so locally reasonable actions can still reduce long-term profit or amplify inventory fluctuations.

The course code provides the base Beer Game environment and a random policy interface. In this project, firm 1 is the learning agent and the other firms use a base-stock heuristic, resulting in a single-agent reinforcement learning problem. Starting from a DQN baseline, we add Double DQN and PPO, and then extend the scalar order action into a three-dimensional order vector to study a harder high-dimensional action space.

This report reorganizes the full current experiment set instead of reporting only partial results. We ask which algorithm is best in the standard action space, how much harder the high-dimensional formulation is, whether PPO is more suitable than DQN-style methods in that setting, and which result should be selected for the final paper.

2. Environment and Task

The environment contains $N = 3$ firms. Firm 0 is closest to the external market, firm 1 is the learning firm, and firm 2 is the upstream supplier. Each episode lasts 100 steps. External demand follows

$$D_t \sim \mathrm{Poisson}(10). \tag{1}$$

The local observation is the firm's own order, satisfied demand, and inventory:

$$s_{i,t} = [q_{i,t}, d_{i,t}, I_{i,t}]. \tag{2}$$

Inventory evolves as

$$I_{i,t+1} = I_{i,t} + q_{i,t} - d_{i,t}. \tag{3}$$

The immediate reward combines sales revenue, procurement cost, holding cost, and shortage penalty:

$$r_{i,t} = p_i d_{i,t} - p_{i+1} q_{i,t} - h I_{i,t} - c \max(0, D_{i,t} - d_{i,t}). \tag{4}$$

The experiments use prices $[10, 9, 8]$, holding cost $h = 0.5$, shortage penalty $c = 2.0$, and initial inventory 100.

3. Algorithms and Action Spaces

3.1. Standard Action Algorithms

In the standard action space, the learning firm selects $a_t \in \{1, \dots, 20\}$ at each step. DQN approximates $Q(s, a)$ with a neural network trained from replay-buffer samples and a target network. Double DQN selects the next action with

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the online network and evaluates it with the target network to reduce overestimation. PPO uses an actor-critic network, GAE, and a clipped policy objective to directly optimize a stochastic policy. The current PPO configuration uses learning rate 10^{-4}, clip range 0.15, entropy coefficient 0.03, and reward scaling 0.01.

3.2. High-Dimensional Action Algorithms

The high-dimensional action space replaces the scalar order with a three-dimensional order vector. Each channel can choose from $\{0, 5, 10, 15, 20\}$, and the total order is constrained to lie between 5 and 20. After invalid combinations are filtered out, the action set contains 34 discrete vector actions. The environment applies channel cost multipliers $[1.0, 1.1, 1.25]$ when computing procurement cost. High-Dim DQN and High-Dim Double DQN learn Q-values over the 34 composite actions. High-Dim PPO treats the composite actions as categories in a categorical policy and maps the selected category back to an order vector.

This enumerated design makes the comparison with DQN straightforward, but it is also difficult: similar order vectors are treated as unrelated actions, and the learner cannot directly exploit structure such as similar total order quantities or channel substitutions. The high-dimensional experiment therefore tests both action-space scaling and exploration quality.

4. Experimental Design

All experiments use Hydra configuration files. Outputs are saved under results/<algorithm>/<timestamp>/ and include the model, training scores, test scores, test trajectories, and `summary.json`. The complete single-run comparison uses the latest available runs in the current repository: standard DQN, Double DQN, PPO, High-Dim DQN, High-Dim Double DQN, and High-Dim PPO are each trained for 2000 episodes and evaluated for 20 test episodes.

A single run is not enough to support a robust conclusion. Different algorithms use different seed offsets, and each algorithm consumes a different amount of randomness during training, so test demand sequences may not be identical. We therefore add a controlled high-dimensional experiment. High-Dim DQN, High-Dim Double DQN, and High-Dim PPO are run with shared seed offsets $\{0, 100, 200\}$, each for 2000 training episodes. Before evaluation, the random stream is reset so that algorithms with the same seed offset face the same test demand sequence.

Table 1. Standard scalar-action performance over 20 test episodes.

Metric	DQN	Double DQN	PPO
Mean reward	611.68	611.88	**627.90**
Reward std.	60.96	48.61	**39.30**
Avg. inventory	18.04	**16.69**	17.41
Avg. order	8.63	**8.20**	8.45
Demand satisfaction	**0.937**	0.917	0.934

Table 2. Three-dimensional vector-action performance over 20 test episodes.

Metric	HD DQN	HD Double DQN	HD PPO
Mean reward	357.75	299.50	**406.63**
Reward std.	100.91	104.83	**74.12**
Avg. inventory	21.73	**20.54**	21.51
Avg. order	8.43	**8.06**	8.42
Demand satisfaction	0.917	0.904	**0.930**

5. Complete Single-Run Results

Table 1 shows that PPO is best in the standard scalar action space, with mean reward 627.90 and the lowest reward standard deviation, 39.30. DQN has a slightly higher demand satisfaction rate, but lower reward, indicating that service level alone does not determine profit because holding and procurement costs also matter. Double DQN reduces variance and inventory relative to DQN, but its mean reward is nearly tied with DQN.

Table 2 shows that High-Dim PPO is the best method in the complete high-dimensional single-run experiment. It obtains mean reward 406.63 and the lowest reward variance. High-Dim DQN reaches 357.75, while High-Dim Double DQN reaches 299.50. This suggests that directly optimizing a stochastic policy is more stable than value-based off-policy learning in the current enumerated vector action space.

Figure 1 visualizes the six-algorithm test comparison: standard PPO is best overall, and High-Dim PPO is best among the high-dimensional agents. Figure 2 shows that scalar-action methods stabilize faster, while high-dimensional methods incur larger early exploration costs.

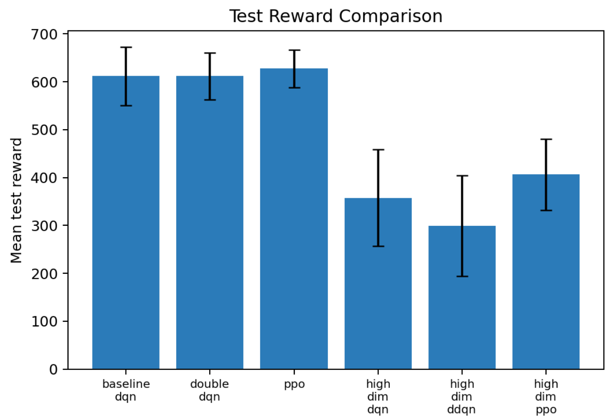

Figure 1. Test reward comparison for the complete single-run experiments, including all six standard and high-dimensional algorithms.

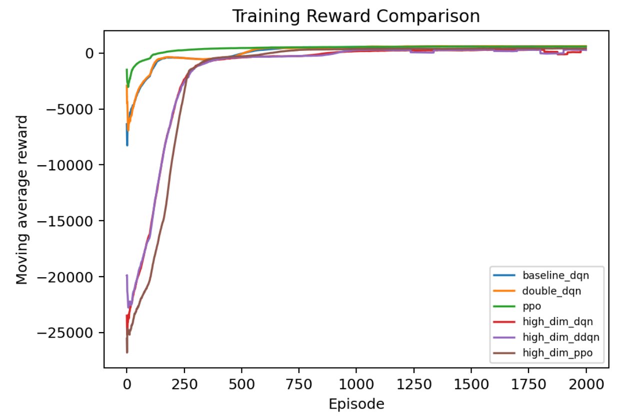

Figure 2. Moving-average training reward curves for the complete single-run experiments. The high-dimensional methods have larger early negative returns, reflecting harder exploration in the composite action space.

6. Controlled High-Dimensional Results

Table 3. Controlled high-dimensional experiment. Means and standard deviations are computed across 3 training seeds; each seed-level value is a mean over 20 test episodes.

Method	Reward	Sat.	Inv.	Order
HD PPO	**407.18±21.33**	**0.946**	**22.25**	8.43
HD DQN	339.91±18.37	0.934	22.76	**8.32**
HD DDQN	306.04±46.72	0.938	23.07	8.35

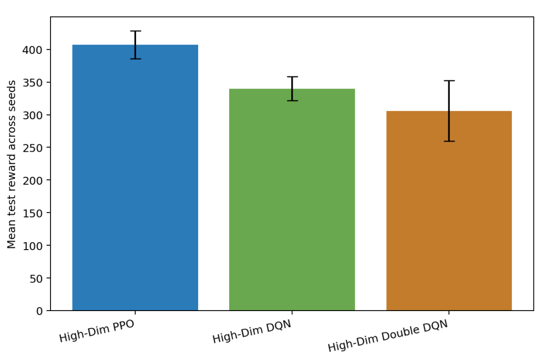

Figure 3. Mean test reward in the controlled high-dimensional 3-seed experiment. Error bars show standard deviation across training seeds.

Table 3 and Figure 3 confirm the single-run result. High-Dim PPO reaches 407.18 ± 21.33 mean reward across three training seeds, outperforming both High-Dim DQN and High-Dim Double DQN. High-Dim Double DQN has the largest cross-seed standard deviation, suggesting higher sensitivity to initialization and exploration path. The best High-Dim PPO controlled run uses seed offset 200 and reaches mean reward 424.95 with demand satisfaction 0.949. We select this High-Dim PPO run as the best high-dimensional policy result.

7. Discussion

The full results show that scalar-action policies still outperform the current high-dimensional policies. This does not imply that vector orders are useless; rather, the current high-dimensional implementation enumerates 34 vector orders as unstructured categorical actions. This increases exploration difficulty and prevents the learner from exploiting relationships between similar order vectors.

Within the high-dimensional task, PPO is the strongest choice. First, in the complete single-run experiment, High-Dim PPO obtains both the highest mean reward and the lowest reward variance. Second, in the controlled 3-seed comparison, PPO remains consistently ahead. Third, PPO's stochastic policy and entropy regularization help maintain exploration over order allocation choices, while DQN-style methods are more exposed to noisy Q-value estimates in the composite action space.

The standard-action results also show that maximizing profit is not the same as maximizing demand satisfaction. Standard DQN has the highest satisfaction rate, but PPO obtains higher reward and lower variance. Future high-dimensional methods should therefore model the tradeoff between procurement, holding, and shortage costs more structurally. A factorized or autoregressive policy that generates orders channel by channel would likely be better than enumerating every vector order as an unrelated category.

8. Conclusion

This report provides a complete comparison of standard and high-dimensional reinforcement learning methods for Beer Game inventory control. In the standard action space, PPO obtains the best mean test reward, 627.90, and the lowest reward variance. In the high-dimensional order space, High-Dim PPO obtains 406.63 in the complete single-run experiment and 407.18 ± 21.33 in the controlled 3-seed experiment; the best controlled PPO run reaches 424.95. These results outperform High-Dim DQN and High-Dim Double DQN. We therefore select High-Dim PPO as the best high-dimensional experimental result and the main method for the final paper.